

Table 1: Effect of Ownership Change on Nursing Staffing

Outcome	RN	LPN	CNA	Total
HPRD	−0.0368*** (0.0055)	−0.0039 (0.0062)	−0.0532*** (0.0096)	−0.0940*** (0.0134)
Log(HPRD)	−0.1174*** (0.0168)	−0.0022 (0.0095)	−0.0270*** (0.0049)	−0.0293*** (0.0041)
Hours	−80.0180*** (15.3077)	68.4756*** (17.8010)	64.8807* (34.9039)	53.3383 (49.9070)
Log(hours)	−0.0767*** (0.0173)	0.0384*** (0.0103)	0.0133* (0.0068)	0.0110* (0.0060)

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period. The sample excludes facilities government-owned at any point during the study period.

HPRD is hours per resident day; hours are the HPRD numerator, reported to test whether the HPRD decline is mechanically driven by the occupancy-rate increase in the denominator.

The anticipation window ($\tau = -3, -2, -1$) is excluded.

Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 2: Effect of Ownership Change on Business-Model Outcomes

Outcome	Coefficient (SE)	Observations
Occupancy rate	2.6841*** (0.3497)	550,330
Spare capacity	-0.0263*** (0.0035)	550,330
Medicare share	1.6100*** (0.2159)	550,330
Medicaid share	-1.1710* (0.5952)	550,330
Average length of stay	-13.3641*** (4.3491)	540,346
Case mix	0.0525*** (0.0059)	457,842

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period. The sample excludes facilities government-owned at any point during the study period.

Occupancy rate is residents as a share of available bed-days; spare capacity is unused certified beds as a share of certified beds. Payer shares are shares of patient days. The anticipation window ($\tau = -3, -2, -1$) is excluded.

Case mix is restricted to 2017/10–2023/12. CMS changed its case-mix methodology in October 2017 and January 2024; both breaks shift the level and the cross-sectional spread simultaneously across the whole sample, which facility and period fixed effects do not absorb.

Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3: Effect of Ownership Change on Quality Measures

Outcome	(1)	(2)	Observations
Panel A: Long-stay labor-saving mechanism measures			
Catheter use	−0.1695*** (0.0488)	−0.1711*** (0.0488)	172,671
Anti-psychotic medication use	−0.3362 (0.2218)	−0.3543 (0.2231)	171,358
Anti-anxiety/hypnotic medication use	−0.2321 (0.2019)	−0.2118 (0.2027)	171,854
Panel B: Long-stay resident outcome measures			
Pressure injuries	0.2586 (0.1762)	0.2825 (0.1783)	121,962
Falls with major injury	0.0125 (0.0636)	0.0138 (0.0644)	175,123
Weight loss	0.2042* (0.1040)	0.2080* (0.1034)	170,733
Decline in physical functioning	0.3240 (0.2695)	0.3501 (0.2782)	163,983
Urinary tract infections	−0.3279*** (0.0765)	−0.3181*** (0.0767)	174,381
Panel C: Short-stay measures			
New antipsychotic medication	−0.0187 (0.0432)	−0.0219 (0.0433)	134,020
Improved function	−0.0569 (0.4888)	−0.0880 (0.4889)	88,792

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period. The sample excludes facilities government-owned at any point during the study period.

Column (1) is the baseline specification. Column (2) adds RN, LPN, and CNA hours per resident day as controls. Because staffing is itself affected by ownership change, column (2) is not a preferred estimate of the total effect; it is reported to show whether the quality response is attenuated after conditioning on measured staffing inputs.

Long-stay measures (Panels A-B) and short-stay measures (Panel C) are constructed from different resident populations and are not directly comparable to one another. For every measure, lower values indicate better measured quality.

Pressure injuries is estimated on 2018–2023 only; improved function is estimated on 2017–2022 only. Both measures show near-complete absence outside these windows and are trimmed to the years where they are actually reported rather than treated as full-panel outcomes. Vaccination measures are excluded from this table.

The transition quarter ($\tau = 0$) is excluded. Observations are reported for column (1); column (2) drops facility-quarters with missing staffing.

Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.